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Compositional Non-Face Re-Identification Pressure under Cumulative Vision Releases

Auditing the "leaky bucket" effect in vision benchmarks

Tirth Joshi • **Honggang Wang***

Katz School of Science and Health, Yeshiva University, New York, USA

tjoshi1@mail.yu.edu   honggang.wang@yu.edu

YESHIVA UNIVERSITY
Katz School of Science and Health

Main message. Removing faces does not make identity risk "reset" with each new release. Dataset expansions, checkpoints, embeddings, indices, and metadata accumulate as side information. We define $vRPI_\alpha$, prove that true-posterior pressure is non-decreasing under added releases, and show large empirical pressure growth on face-masked Re-ID benchmarks.

1. Why Single-Release Privacy Is the Wrong Unit

Most vision privacy evaluations inspect a dataset or model *in isolation*. Real ecosystems publish artifacts sequentially: more data, new checkpoints/APIs, embeddings, retrieval indices, and sometimes metadata.

Latent identity and non-face view. Let $U \in \{1, \ldots, N\}$ index identity, X be a raw observation, and

$$\widetilde{X} = \tau(X)$$

be a non-face transformation. Identity can still remain in clothing, body shape, pose/gait, carried objects, and scene context.

The compositional question. Instead of asking whether one release leaks identity, ask how the posterior

$$\Pr(U \mid Z_{1:T})$$

sharpens as releases accumulate. The operational target is Bayes-optimal guessing:

$$P_{\text{guess}}(U \mid Z_{1:T}) = \mathbb{E}\left[\max_u \Pr(U = u \mid Z_{1:T})\right].$$

2. Release-as-Channel Formulation

R1 Sparse exposure encoder E_1 → **R2** Full exposure $+E_2$ → **R3** Generic channel $+E_3$ → **R4** Prototype bank + kNN graph

$Z_{1:T} = (Z_1, \ldots, Z_T)$ **accumulates side information**

↓

$\Pr(U \mid Z_{1:T})$ becomes the object of the privacy audit

Each disclosure is modeled as a channel $Z_t \sim K_t(\cdot \mid \widetilde{X})$. The framework is release-sequence aware rather than tied to one Re-ID model or anonymizer.

3. $vRPI_\alpha$: Metric and Core Theory

Rényi pressure index. For $\alpha > 1$, using Arimoto conditional Rényi entropy,

$$\boxed{vRPI_\alpha(T) = \log N - H_\alpha(U \mid Z_{1:T})}.$$

For a uniform prior, $\log N$ is maximal identity uncertainty, so $vRPI_\alpha$ measures how much anonymity has been eroded by cumulative releases.

Theorem 1: monotonicity on the true posterior

$$H_\alpha(U \mid Z, W) \le H_\alpha(U \mid Z) \Longrightarrow vRPI_\alpha(T+1) \ge vRPI_\alpha(T).$$

Added side information cannot restore identity uncertainty.

Theorem 2: operational guessing bridge

$$P_{\text{guess}}(U \mid Z) \le \exp\left(\frac{1-\alpha}{\alpha} H_\alpha(U \mid Z)\right).$$

Thus higher pressure corresponds to a smaller effective anonymity set and stronger identity inference.

Choosing α. $\alpha = 2$ is the recommended stable default; $\alpha = 5$ or 10 emphasizes high-confidence / worst-case posterior concentration and should be paired with calibration diagnostics.

Additional guarantees.

- **Ideal additivity benchmark.** If releases are conditionally independent given identity, Sibson information factorizes and gives an additive upper benchmark on pressure:

$$vRPI_\alpha(T) \le \sum_{t=1}^{T} I_\alpha^{\mathrm{S}}(U; Z_t).$$

- **Ablation monotonicity.** If τ_B is a post-processing of τ_A, then removing information cannot increase pressure:

$$vRPI_\alpha^A(T) \ge vRPI_\alpha^B(T).$$

- **Plug-in stability.** If the estimated posterior is close in expected ℓ_1 error, the estimated $vRPI_\alpha$ changes continuously, with an explicit logarithmic error bound.

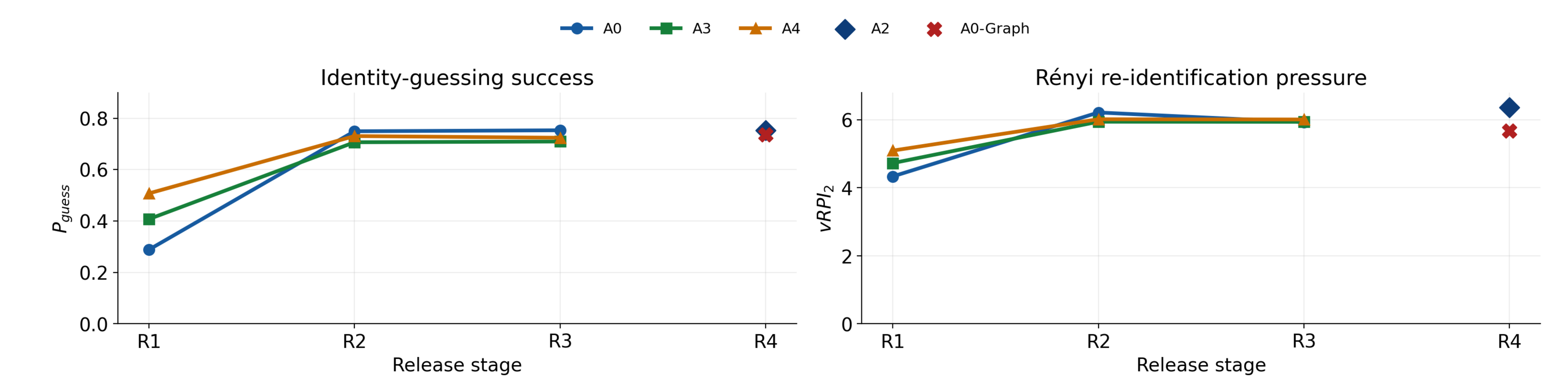
4. Main Empirical Result on Market-1501-τ

The dominant leak is exposure expansion. From R1 to R2, retrieval attacker A0 jumps from $P_{\text{guess}} = 0.288$ to 0.749 and $vRPI_2 = 4.332$ to 6.212. A3 rises from 0.407 to 0.706. R4 then shows that publishable artifacts themselves create exploitable leakage pathways.

Release	Attacker	P_{guess}	vRPI_2	Interpretation
R1	A0	.2883	4.3319	Sparse exposure baseline
R2	A0	.7491	6.2119	Dominant exposure-expansion jump
R3	A0	.7529	5.9271	Generic channel; calibrated fusion
R4	A2	.7519	6.3572	Prototype/index specialist
R4	A0-Graph	.7366	5.6697	kNN-graph pathway

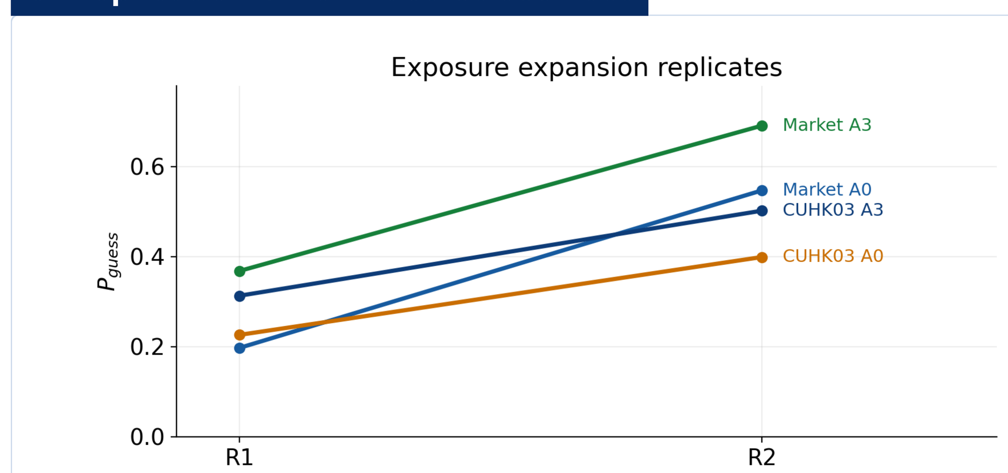
Mean over 3 seeds. A3 independently shows the same R1→R2 pattern (P_{guess}: .407→.706; $vRPI_2$: 4.723→5.940). Empirical plug-in estimates can fluctuate with attacker/calibration even though Theorem 1 is a true-posterior statement.

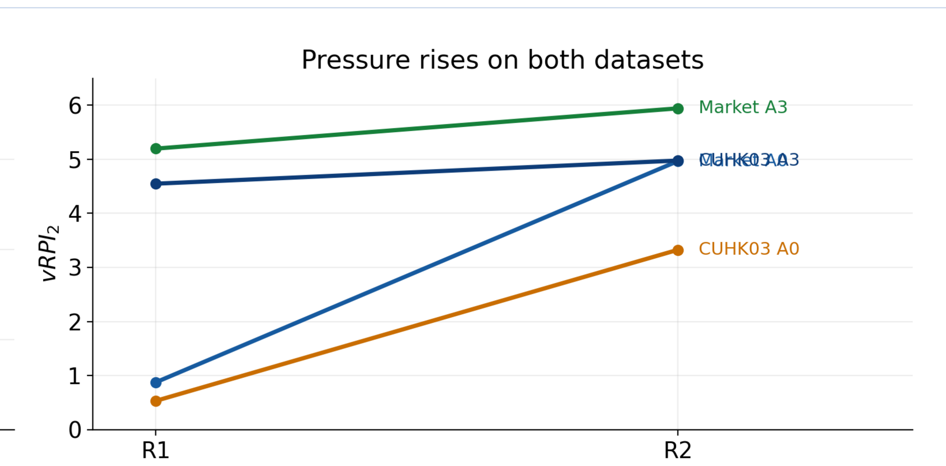
5. Release Trend: The R1→R2 Jump Dominates



The exposure expansion is the clearest compositional effect. R2→R3 is stable under validation-tuned fusion and temperature scaling. R4 demonstrates artifact-specific leakage through the prototype/index specialist and graph-exploiting attacker.

6. Replication + Non-Face Stress Test





CUHK03-τ replication. A lighter paired protocol still yields a positive R1→R2 increase for both retrieval (A0) and nonlinear-probe (A3) attackers on Market and CUHK03. The absolute magnitudes are not compared across protocols; the paired direction is the claim.

Masking ablation. Sweeping top-region masking from 0% to 40% leaves pressure essentially flat: A0 $vRPI_2 \in [4.87, 4.91]$, A3 $vRPI_2 \in [5.83, 5.88]$. Even at 40% masking, $P_{\text{guess}} \ge .51$ (A0) and $\ge .64$ (A3).

Important scope. The top-region mask is a controlled proxy, not a realistic detector-based or generative anonymizer. The result shows persistent non-face signal under aggressive region removal; it does not claim that all anonymization methods fail.

7. Experimental Protocol and Attacker Ladder

Dataset. Main experiment: Market-1501 training identities, $N = 751$, with the top 20% of each image masked (Market-1501-τ). Images are split into exposure, gallery, query, and held-out validation pools over the same closed-world label space.

What counts as a release?

- **R1:** encoder E_1 trained on sparse exposure.
- **R2:** add E_2 trained on full exposure; cumulative feature state.
- **R3:** add generic ImageNet-pretrained ResNet-50 E_3.
- **R4:** publish an identity prototype bank and a kNN graph derived from the R3 gallery.

Attacker ladder.

- **A0** black-box retrieval with identity-level posterior aggregation.
- **A1** linear probe.
- **A2** prototype-index specialist (R4).
- **A3** MLP / nonlinear probe.
- **A4** white-box fine-tuning (R1–R3).

Every attacker maps scores into a posterior over the same identity universe before entropy computation.

8. Diagnostics: Dependence, Calibration, Priors

Dependence overlap. Conditional independence gives an ideal additive benchmark, not a claim about real CV releases. On Market,

$$\Delta_{\text{corr},2} = \sum_t \widehat{vRPI}_2(Z_t) - \widehat{vRPI}_2(Z_{1:T}) = 0.304 > 0,$$

quantifying redundancy among correlated release channels.

Temperature calibration. Across $\theta \in \{.025, .05, .1, .2, .5, 1.0\}$, absolute pressure changes, but

$$\Delta vRPI_2(R2 - R1) > 0$$

throughout: A0 spans $[.781, 4.100]$ and A3 $[.567, .677]$.

Non-uniform identity priors. For a Zipf(s) prior, normalize by prior entropy:

$$vRPI_\alpha^\pi(T) = H_\alpha(U) - H_\alpha(U \mid Z_{1:T}).$$

The R1→R2 increment shrinks from 3.818 at $s = 0$ to $.123$ at $s = 1.5$, because a concentrated prior leaves less anonymity to erode.

9. What the Framework Adds

Not another Re-ID model. $vRPI_\alpha$ audits the privacy pressure induced by releases; rank/mAP remains a retrieval-utility metric. The two answer different questions.

Not a replacement for maximal leakage or DP. Maximal leakage optimizes adversarial gain at the channel level, while DP supplies worst-case stability guarantees. $vRPI_\alpha$ instead fixes the vision identity task and measures posterior concentration over a release sequence.

Governance implication. Privacy review should happen at the **sequence level**: evaluate planned dataset/model/index disclosures together, minimize unnecessary metadata, and treat high-leverage artifacts such as embedding dumps, prototype banks, and indices as privacy-relevant releases.

Positioning against common criteria.

Measure	Object measured
Rank/mAP	Retrieval ranking quality
P_{guess}	Bayes posterior maximum
$vRPI_\alpha$	Rényi posterior concentration
Maximal leakage	Worst-case channel gain
DP / attribute inference	Worst-case stability / bound

Release hygiene, not surveillance. The intended use is governance: audit successive disclosures before publication, limit high-leverage artifacts, and minimize unnecessary metadata exposure.

Limitations.

- Closed-world image benchmarks; broader datasets, video, larger scale, and open-set evaluation remain future work.
- Top-region masking is a controlled proxy rather than realistic anonymization.
- $vRPI_\alpha$ is an audit statistic, not a mitigation method.

10. Final Contribution, Empirical Signal, and Limitation

Contribution

A **release-sequence privacy formalism** and $vRPI_\alpha$, derived from Arimoto conditional Rényi entropy.

Theory: monotonicity under added side information, a Bayes-guessing bridge, an ideal additive benchmark, ablation monotonicity, and plug-in stability.

Empirical signal

On face-masked Market-1501, **exposure expansion causes the dominant jump**: A0 P_{guess} rises .288→.749 and $vRPI_2$ rises 4.332→6.212.

CUHK03 preserves the paired R1→R2 direction, while prototype/index and kNN-graph releases create artifact-specific leakage pathways.

Limitation

The evaluation is intentionally theory-aligned and closed-world. $vRPI_\alpha$ is an **audit statistic, not a mitigation**.

Next: **open-set posteriors, realistic anonymizers, more datasets/video, and longer real release sequences** that better match evolving benchmark ecosystems.